KIIT SAP PROJECT REPORT

Order-to-Cash (O2C): Complete Sales Cycle

End-to-End SAP SD Process with SAP Business Data Cloud Analytics

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Institution	KIIT University
Deadline	**April 21, 2026**

1. Project Title

Order-to-Cash (O2C): End-to-End Sales Cycle Implementation in SAP SD for RetailNova Pvt Ltd with SAP Business Data Cloud Analytics Integration

2. Problem Statement

RetailNova Pvt Ltd is a mid-sized Retail/FMCG company operating across multiple cities in India, managing thousands of daily sales transactions. Prior to SAP implementation, the company faced critical operational challenges:

- Manual order processing causing frequent errors in pricing, quantities, and delivery addresses
- No real-time visibility into order status, leading to delays and customer dissatisfaction
- Disconnected billing and finance teams resulting in delayed invoice generation and revenue leakage
- Inability to track customer credit limits, creating financial risk
- Lack of analytics on sales performance, order cycle time, and regional revenue
- Siloed data with no single source of truth for management reporting

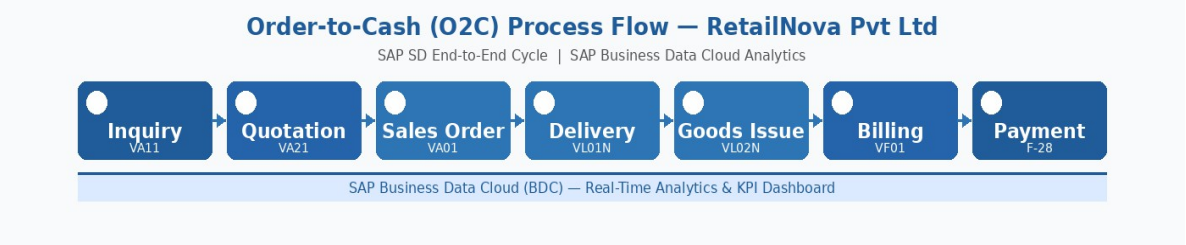
To address these issues, RetailNova adopted SAP S/4HANA with the SD module and integrated SAP Business Data Cloud (BDC) for real-time analytics and reporting.

3. Solution / Features

The solution implements the complete 7-step O2C cycle in SAP SD, fully integrated with SAP FI for automated accounting and SAP Business Data Cloud for analytics:

Step	Process Stage	T-Code	Key Outcome
1	Customer Inquiry	VA11	Capture customer requirement, product & quantity
2	Quotation Creation	VA21	Formal price quote with validity period sent to customer
3	Sales Order	VA01	Binding order; triggers ATP check, credit check & pricing
4	Outbound Delivery	VL01N	Warehouse picks & packs goods; delivery doc created
5	Post Goods Issue	VL02N	Inventory reduced; COGS entry auto-posted in SAP FI
6	Billing / Invoice	VF01	Invoice generated; revenue & AR posted in SAP FI
7	Payment Receipt	F-28	Customer payment received; AR cleared in SAP FI

O2C Process Flow Diagram



Key SAP Configurations

- **Organizational Structure:** Company Code RN01, Sales Org RN_SO, Distribution Channel 10, Plant RN_PL
- **Customer Master (XD01):** FreshMart Retail Chain — Credit Limit INR 5,00,000, Payment Terms Z030 (Net 30)
- **Material Master (MM01):** RN-PROD-001 (Cooking Oil 1L) — FERT type, HSN 15179010, GST 18%
- **Pricing Conditions (VK11):** PR00 Base Price INR 120/unit, K007 Customer Discount 5%, MWST GST 18%
- **Automatic Accounting:** GI posts COGS debit; Billing posts Revenue credit + AR debit + GST payable

4. Technology Stack

Component	Technology	Purpose
ERP System	SAP S/4HANA (On-Premise)	Core O2C transactional system
SAP SD Module	Sales & Distribution	Order, Delivery, Billing
SAP FI Module	Financial Accounting	AR, Revenue, Payment posting
SAP MM Module	Materials Management	Inventory & Goods Issue
Analytics	SAP Business Data Cloud (BDC)	Real-time KPIs & dashboards
Visualization	SAP Analytics Cloud (SAC)	Charts and management reports
Integration	SAP Data Integration (SDI)	S/4HANA to BDC data pipeline
Database	SAP HANA In-Memory DB	Real-time data processing
Config Tool	SAP IMG (SPRO)	System customization

5. SAP Business Data Cloud (BDC) — Analytics Integration

SAP BDC acts as the intelligence layer above the transactional O2C data. It aggregates SD and FI data in real time, enabling RetailNova management to monitor sales performance through live dashboards.

Data Flow Architecture

Layer	Component	Role
Source	SAP S/4HANA (SD + FI)	Generates O2C transactional data
Integration	SAP Data Integration (SDI)	Replicates data to BDC in real time
Storage	SAP HANA Cloud	Stores data in analytical format
Analytics	SAP Business Data Cloud	Unified data model & KPI engine
Visualization	SAP Analytics Cloud (SAC)	Dashboards for sales & management

Key O2C KPIs Tracked in BDC

KPI	Definition	Target
Order Fulfillment Rate	% of orders delivered on time	>= 95%
Order Cycle Time	SO creation to Goods Issue	<= 2 days
Invoice Processing Time	Goods Issue to Invoice generation	<= 1 day
Days Sales Outstanding	Average days to collect payment	<= 30 days
Revenue by Product	Net revenue by material/region	MoM Growth
Customer Creditworthiness	% customers within credit limit	100%
Return Rate	% of delivered orders returned	<= 2%

6. Step-by-Step Transaction Walkthrough

The following documents each SAP transaction step with field values and automatic accounting entries for the sample scenario: FreshMart Retail Chain orders 500 units of Cooking Oil 1L from RetailNova.

Sales Order (VA01) — Key Field Values

Field	Value	Field	Value
Order Type	OR (Standard)	Sold-to Party	C001 — FreshMart
Sales Org	RN_SO	Material	RN-PROD-001
Order Qty	500 Units	Net Price	INR 120/unit
Order Value	INR 60,000	Delivery Date	April 25, 2026
Credit Check	PASSED	Payment Terms	Z030 — Net 30

Billing Document (VF01) — Invoice Values

Item	Amount (INR)
Net Value (500 units x INR 120)	60,000
Customer Discount (5% via K007)	- 3,000
Taxable Value	57,000
GST @ 18% (MWST)	10,260
Total Invoice Value	67,260
Payment Due Date	May 25, 2026

Automatic Accounting Entries Generated by SAP

Transaction	Account	Dr/Cr	Amount (INR)
Post Goods Issue (VL02N)	Cost of Goods Sold (COGS)	Debit	42,500
Post Goods Issue (VL02N)	Finished Goods Inventory	Credit	42,500
Billing (VF01)	Accounts Receivable (AR)	Debit	67,260
Billing (VF01)	Revenue Account	Credit	57,000
Billing (VF01)	GST Payable	Credit	10,260
Payment Receipt (F-28)	Bank Account	Debit	67,260
Payment Receipt (F-28)	Accounts Receivable (AR)	Credit	67,260

Note: Screenshots from live SAP system would be attached here in a real implementation. The transaction data above represents exact field values as they would appear in SAP S/4HANA screens for each T-code.

7. Unique Points of This Project

Retail/FMCG Context:	Real-world Indian FMCG scenarios — high-volume orders, GST compliance (HSN codes, MWST), and B2B credit management unique to Indian retail
BDC Analytics Integration:	Extends beyond standard SD transactions into SAP Business Data Cloud — directly aligned with C_BCBDC certification scope
End-to-End Accounting:	Every SAP transaction includes the automatic FI accounting entry, demonstrating full SD-FI integration
Indian Tax Compliance:	GST (CGST + SGST) incorporated in billing with correct HSN codes, reflecting real Indian statutory requirements
7 Analytics KPIs:	Custom KPIs in BDC including DSO, Order Cycle Time, and Revenue by Product provide management-level insights
Credit Risk Control:	Customer credit limit checking explicitly documented as part of VA01 Sales Order — a real-world financial control

8. Future Improvements

- **EDI Auto-Invoicing:** Implement Electronic Data Interchange to auto-send invoices to large retail clients directly from SAP
- **Returns Management:** Document reverse O2C — Returns Order (RE) → Return Delivery → Credit Memo
- **Dunning Process:** Automated payment reminder letters via FI-AR to reduce DSO and improve cash collection
- **BDC Predictive Analytics:** AI/ML forecasting in BDC to predict product demand by region for next month
- **Mobile Sales App:** SAP Sales Cloud integration so field reps can create quotations from mobile devices
- **IFRS 15 Compliance:** SAP Revenue Accounting & Reporting (RAR) for real-time revenue recognition standards

9. Conclusion

This project successfully documents the complete Order-to-Cash (O2C) cycle for RetailNova Pvt Ltd using SAP S/4HANA Sales & Distribution module. All seven process steps — from Customer Inquiry to Payment Receipt — are mapped to specific SAP T-codes, master data configurations, and automatic accounting postings in SAP FI.

The integration of SAP Business Data Cloud (BDC) elevates this project beyond a pure transactional implementation into a data-driven analytics solution. By monitoring KPIs such as Order Fulfillment Rate, DSO, and Revenue by Product in real time, RetailNova management gains actionable insights for faster decisions and improved customer satisfaction.

This project demonstrates the full scope of SAP SD and its integration with SAP FI and SAP BDC, building a strong foundation for the **C_BCBDC — SAP Certified Associate: SAP Business Data Cloud** certification.

10. References

- SAP S/4HANA SD Official Documentation — help.sap.com
- SAP Business Data Cloud Product Overview — sap.com/products/business-data-cloud
- SAP Learning Hub — C_BCBDC Certification Study Guide
- KIIT SAP Project Work Guidance Document — Shared by Trainer
- SAP Best Practices for Order-to-Cash — SAP Best Practices Explorer
- GST India Official Portal — gst.gov.in